

Milestone Two: Global Climate Impact on US Agriculture

DX799 01 Data Science Capstone

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Problem Statement:

Climate change is a serious and complex problem that has been plaguing the world since the inception of the industrial age. Humanity has exploited and produced emissions from carbon sources since early times, but has massively been accelerating our use of these resources since the 1700s (US Department of Energy, 2026). Coal, Oil, and Gas make up the majority of power generation, even in developed countries (International Energy Authority, 2025). While the process of moving to renewable energy has begun, the transition has been slow and not necessarily an important issue to most world leaders. The geopolitics of propping up the fossil fuel industry have been stifling innovation in that area, and its impacts have the potential to be catastrophic.

The ramifications of fossil fuel usage is a slow, but noticeable, change in global climate. Temperatures are on the rise, catastrophic weather events are increasing in most parts of the world, and industries have been impacted negatively as a result. Agriculture in specific is extremely dependent on climate for crop yield and drought mitigation. While we are unsure to what extent the climate is repairable, we can explore the impacts of CO₂ emissions on agriculture.

Modeling Techniques: Breadth

The first technique utilized for analysis was K-nearest Neighbors (KNN). This analysis is a simple machine learning algorithm that groups data by looking at the closest neighbors. For numeric datasets, this method predicts the average numeric value for a given data point based on the nearest neighbors. For this analysis, datasets were separated into target and features. Data was then split into testing and training sets utilizing test-train-split. Features were then scaled using standard scaler to standardize the analysis and improve accuracy and efficiency of the

model. A KNN model was then fitted utilizing 5 neighbors ($k=5$) to test the various distance metrics available: Euclidian, Manhattan, Minkowski, and Cosine. Then, the metric that performed the best (based on minimized MSE, RMSE, and MAE and maximized R^2) was tuned for the optimal K value. For the Agriculture dataset, the Manhattan distance was found to be the most accurate at $K=5$. The optimal K value is 20, according to my analysis. This is a relatively high K value. K values were tested until $K=40$ and there was not another local maximum that performed better than $K=20$. The results of the K tuning are shown in Figure 1. For the CO2 dataset, the Manhattan distance was once again found to be the optimal distance metric. The optimal K value was found to be $K=1$. A low K value is indicative of overfitting, which would follow with other observations about this dataset. Previous analysis has shown that this dataset has high correlations between the features and the target.

The next analysis technique utilized was K-Means. This differs from KNN in that K-means is an unsupervised learning model that breaks data into groups, rather than predicting the value of new data based on the surrounding data. This model was utilized to see if there are any groups that may be observed within the dataset that may not be apparent through simpler analysis such as linear regression. Data was scaled using a standard scaler. This is imperative for K-means analysis, and will be further discussed in the overfitting section of the paper.

Two metrics were observed using K-means to calculate the optimal number (k) of groups, the elbow method and the silhouette score. FIG. 3 shows the optimization of both of these metrics for the Agriculture dataset. The analysis shows that $K=2$ is the optimal K value for K-means.

Hierarchical Agglomerative Clustering (HAC) was the next technique utilized to analyze my datasets. This technique clusters data based on similarities, computes the distance between the clusters, merges the two closest clusters, and repeats this process until one cluster remains.

This analysis can use different linkage methods: Single linkage, complete linkage, average linkage, and ward linkage, and uses the same distance metrics as KNN: Euclidean, Manhattan, Cosine, etc. HAC is usually visualized as a dendrogram, which shows the sequence of the cluster merges. The leaves show the individual data points, and the branches show the merges and the height of a merge represents the distance between the clusters. You can truncate the dendrogram to show a subset of clusters for readability, especially with large datasets. Figures 3 and 4 show the dendrograms for the CO2 dataset and the Agriculture dataset respectively. Prior to dendrogram generation, both datasets were tested with each linkage type, and ward linkage performed the best for both data sets. FIG. 3 shows that the CO2 dataset has three distinct clusters that make up the dendritic tree, denoted in blue, orange, and green. The Agriculture dataset, shown in FIG. 4 has also been truncated to three distinct clusters. This truncation helps make the data more readable, and thus easier to show in a dendrogram. These analyses show that there is one cluster from the dataset that is very different from the rest of the data, and two clusters that have more similarities among themselves.

The final analysis performed on the data is Density-Based Spatial Clustering of Applications with Noise (DBSCAN). This technique labels clusters based on density. The two values needed to tune for DBSCAN are epsilon, which tunes the radius of a neighborhood, and minpts, which determines the minimum number of points required to consider it a core point. StandardScaler was used to standardize data before analysis. DBSCAN was utilized on each feature against the dataset target for both datasets. FIG. 5 shows an example of the DBSCAN model for the CO2 dataset using the CO2 per GDP feature. The core, border, and noise points are labelled in green, yellow, and red respectively. This shows that some data points can be looked at with a grain of salt, as the data labelled noise may not be representative of the trend that could be

gleaned from the dataset. This analysis helps further refine where the trend within the data may be.

Modeling Techniques: Depth

The analysis technique to explore in depth is Gradient Boosting Analysis. This analysis utilizes weak decision trees, known as weak learners, that are built sequentially and corrects errors from previous trees. The parameters that need to be tuned are number of trees, learning rate, and depth. Feature importance can also be calculated in order to determine which features contribute the most to improving the model. For both of the datasets, a baseline gradient boosting model was run using default parameters. Data was preprocessed using test train split. Hyperparameters were tuned using loops, iterating on the model with different values for the key parameters and comparing the mean squared error. The hyperparameter tuning and feature importance is shown in FIG. 6 and FIG.7.

For FIG 6 and the CO2 dataset, it was found that the optimal number of trees was around 100. The most important feature for predicting temperature change from CO2 was GDP. The optimal learning rate was 0.1, and the best max depth was found to be 1. The tuned parameters were then used to re-run the model and the resulting MSE, MAE, and R^2 were compared to the baseline model. The baseline values were: MSE 0.0011, MAE 0.0051, and R^2 0.7773. The optimized values were MSE 0.0011, MAE 0.0088, and R^2 0.7898. This shows an improvement in R^2 value by about 1.61%. This improvement is minor, and it seems that the baseline test was performing well for the CO2 dataset.

The agriculture dataset, shown in FIG.7 was found to show optimal values in the following: number of trees: 100, Learning rate: 0.1, Max Depth: 1. The most important feature for predicting Crop Yield was Economic Impact, followed by Average Temperature. The same

baseline metrics were compared to the tuned model. The baseline values were: MSE 0.4185, MAE 0.5320, and R^2 0.5518. The optimized values were MSE 0.3851, MAE 0.5098, and R^2 0.5876. This showed a MSE decrease of 7.97%, a MAE decrease of 4.17%, and a R^2 increase of about 6.49%. This is a significant difference between the baseline and tuned models, which makes sense as there is more noise and variability within the agricultural dataset compared to the CO2 dataset.

The result of the agricultural dataset made me interested in dropping the Economic Impact variable so that I could see the true effect of Average Temperature on predicting Crop Yield. FIG. 8 shows the agricultural dataset predicting crop yield without the economic impact column. Baseline and tuned metrics were compared for this new analysis. The baseline values were: MSE 0.7452, MAE 0.7362, and R^2 0.2020. The optimized values were MSE 0.7241, MAE 0.7364, and R^2 0.2245. This showed a MSE decrease of 2.83%, a MAE increase of 0.03%, and a R^2 increase of about 11.14%. This is a massive increase in R^2 which shows the importance of hyperparameter tuning. This also shows that the temperature and farming methods can indeed be used to predict crop yield, but at a much higher error rate than if the economic impact is included in the analysis.

Overfitting

A train-test split was used to evaluate how well the model generalized to unseen data and to reduce the risk of overfitting. The training set was used to fit the model, while the test set was withheld until evaluation. This prevents the model from being assessed only on data it has already learned from. Comparing training and test performance helps identify overfitting: if the model performs much better on the training data than on the test data, it may be memorizing patterns rather than learning relationships that generalize. This method was appropriate because

the datasets were large enough to split while still maintaining sufficient training data. Compared with cross-validation, a train-test split is less computationally expensive for large datasets, while regularization is a useful but model specific technique for controlling overfitting.

StandardScaler was used to standardize the numeric features before modeling. While standardization does not directly prevent overfitting, it improves model stability and ensures that features with larger numeric ranges do not dominate the learning process. This is especially important for distance-based methods.

To reduce overfitting in the KNN model, the number of neighbors was tuned so that predictions were not based too heavily on individual observations. Train-test splitting was used to evaluate performance on unseen data and guide hyperparameter selection. Feature selection (performed in previous Milestones) can also help reduce overfitting by removing irrelevant variables that distort distance calculations.

Metrics and Hyperparameter Tuning

As discussed in the depth section, the metrics that were used for this analysis were MSE, MAE, and R^2 . MSE and MAE were utilized to determine error. MSE measures the average squared difference between predicted and actual values, giving greater weight to larger errors. MAE measures the average absolute difference between predicted and actual values, providing a straightforward measure of prediction error. R^2 measures the average absolute difference between predicted and actual values, providing a straightforward measure of prediction error.

Hyperparameter tuning was used in most of the models, using the metrics above to determine the optimal values of each parameter. KNN, K-means, and gradient boost were all tuned for hyperparameters, plugging in multiple values for the variables needing to be tuned

(discussed above) and selecting for the best ones. This showed a marked improvement from the baseline models in all three of the models. HAC was also tuned for the best linkage method.

Exploratory Data Analysis

Previous Exploratory Data Analysis (EDA) in earlier milestone projects led to removal of variables that did not correlate with target variables. This was performed using heat maps. MSE was chosen as a metric due to the lack of extreme outliers in the dataset. The Economic Impact column was removed from the gradient boost analysis in order to see the true correlation between Average Temperature and Crop Yield. Economic Impact would be correlated with crop yield, as more crops mean more money, but would not show the impacts of climate change on agriculture. This strengthens my data's correlation with the research question.

Expected VS Unexpected Results

The results from this analysis are promising. I think that the gradient boosting model showing that the agriculture dataset can be predicted from the temperature and agriculture metrics is really interesting. Perhaps crop yield truly is impacted by climate events. Although climate events have lots of noise, there are correlations beyond just the economic impact of the crops. I also found it interesting that GDP is so strongly correlated with CO2 production in the CO2 dataset. It outperforms anything else when predicting temperature increase from CO2 emissions. This proves that countries that are rich are largely responsible for increase in atmospheric temperature. Ironically, these same countries have the greatest access to green technologies. Perhaps, if rich countries were to incentivise green tech more, overall temperature increase would lessen.

Figures

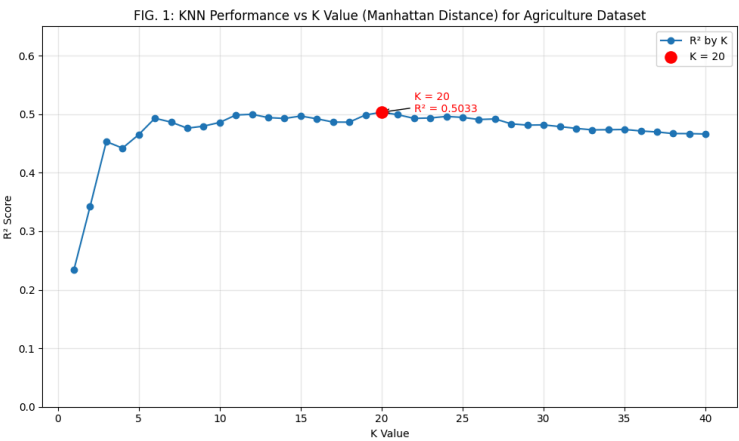


FIG. 1: KNN K value optimization for Agriculture Dataset

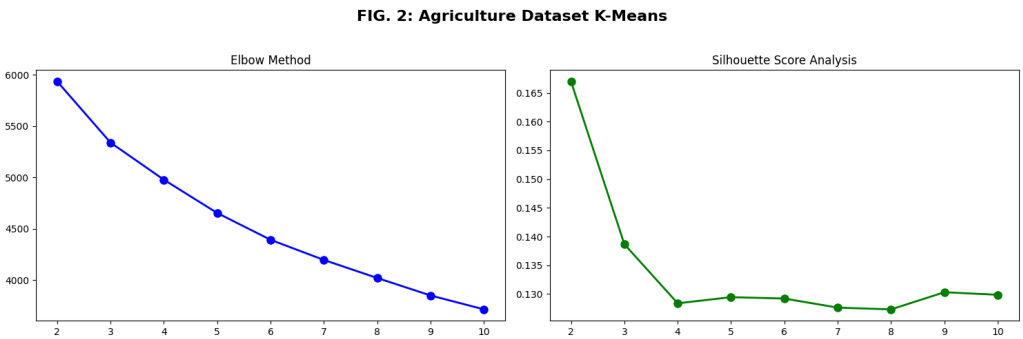


FIG. 2: K-Means Elbow method and Silhouette

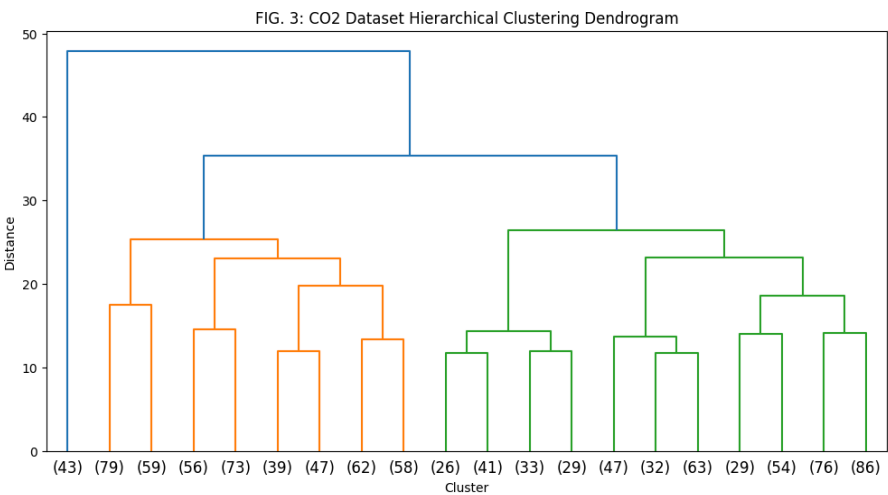


FIG. 3: HAC Dendrogram for CO2 Dataset

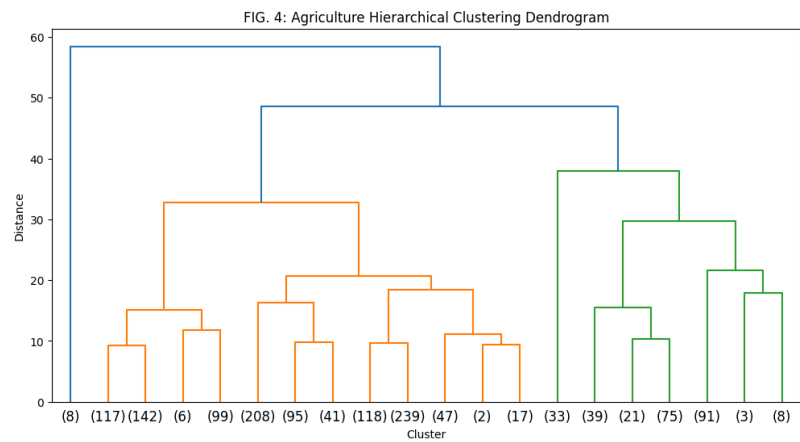


FIG. 4: HAC Dendrogram for Agriculture dataset

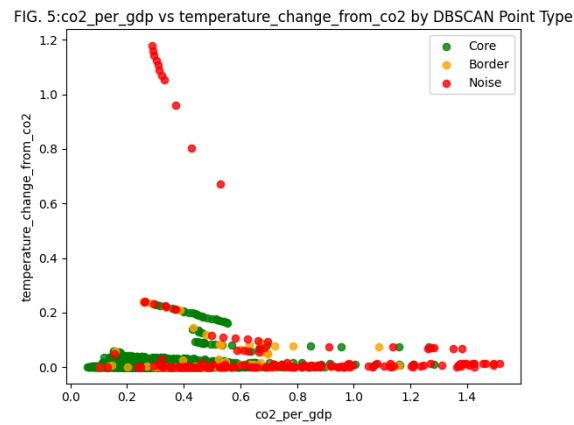


FIG. 5: DBSCAN visualization of CO2 Dataset, CO2 per GDP vs Temperature change from CO2

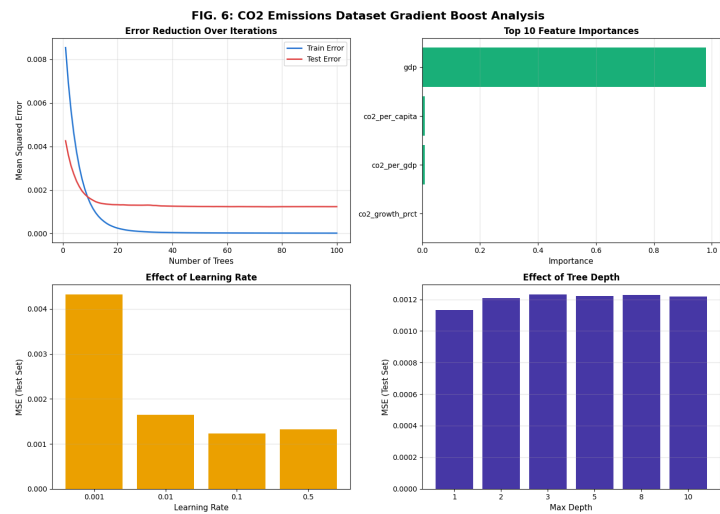


FIG. 6: CO2 Dataset Gradient Boost Analysis

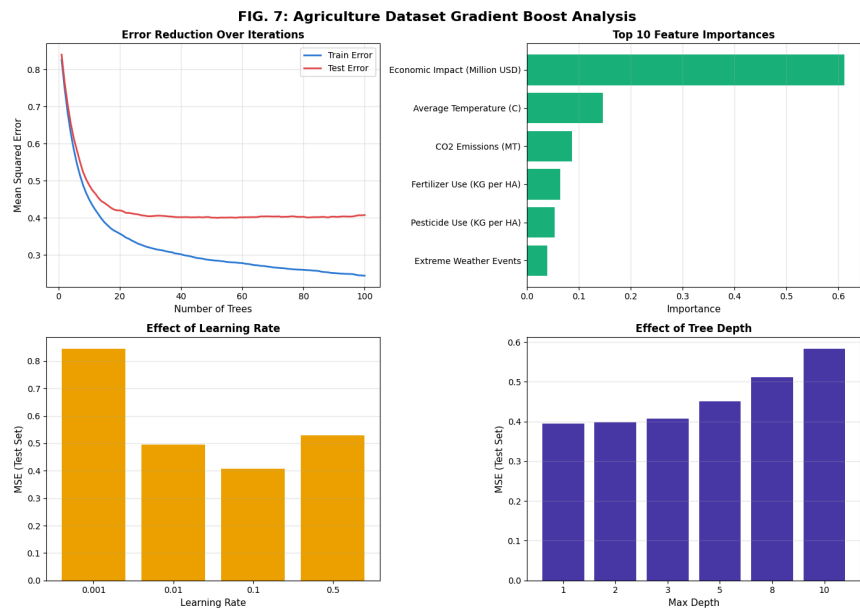


FIG. 7: Agriculture Dataset Gradient Boost Analysis

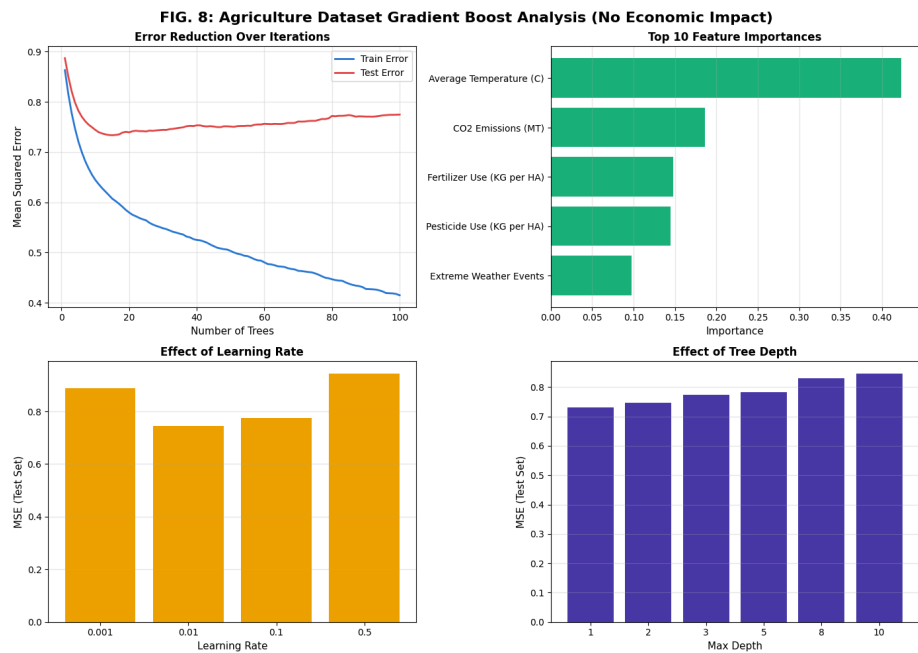


FIG. 8: Agriculture Dataset Gradient Boost Analysis without Economic Impact

AI Appendix

I used AI more than usual in the writing of this paper. I have a concussion so my thinking and focus are impaired, so I thought I needed the extra help. I used AI to both code and assist with analysis and writing. The link to my conversation is here:

<https://terriergpt.bu.edu/share/WaimzWPn9eB0VJXnZl-3j>

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